

# ARCHAI

Sovereign, local-first AI for cultural heritage, turning collections into *living archives* people can talk to, while every record stays on hardware the institution owns and controls.

---

Research, technical overview & collaboration agenda   Rob Graham · rob@fineartmedia.tech  
PhD (Design), RMIT University · FAMTEC

---

## 01 The research

**Cultivating a Living Archive: Designing Sovereign AI Infrastructure for Cultural Heritage.** A practice-based Design PhD at RMIT University, developed through Research through Design, where building the working system *is* the primary act of discovery, not an illustration of theory. A related paper was accepted for the 6th Summit on New Media Art Archiving at ISEA2026, Dubai.

### Primary research question

*How can design transform cultural heritage collections into living archives through sovereign, local-first AI infrastructure while preserving provenance, institutional authority and curatorial practice?*

### Secondary questions

- 1 What design principles support trustworthy conversational engagement with collections while maintaining archival integrity?
- 2 How can sovereign, local-first AI improve accessibility without compromising institutional control of cultural knowledge?
- 3 How might conversation strengthen public engagement while complementing, rather than replacing, curatorial expertise?
- 4 What transferable design framework could support ethical adoption of sovereign AI across museums, galleries, libraries and archives?

---

## 02 What ARCHAI is

Not a new collection system, a **semantic interface layer** that sits *above* whatever an institution already runs (EMu, Axiell, Vernon, TMS, CollectiveAccess, ResourceSpace, and the rest). It connects via their APIs, reads and normalises the metadata, and makes the collection **semantically searchable and conversationally accessible**, no migration, nothing replaced, and it never writes back to the record.

Because it runs on institutionally owned hardware with local-first models, the collection *and* the new layer of visitor conversation it creates stay in the building. Sovereignty here is not just an ethical preference, it is a preservation requirement: an institution that can't maintain independent access to its own interpretive systems has, in a real sense, lost control of its collection.

**The living-archive proposition.** Instead of a static repository visitors must search, the archive becomes a place that can hold a conversation while staying anchored in evidence. Every response is bounded by verified collection information, the archive doesn't become less authoritative, its knowledge becomes more accessible.

---

## 03 Technical overview

### Three-layer architecture

#### HERITAGE FOUNDATION

The institution's CMS /  
DAMS

Read-only. No AI output ever  
enters the record. Permanent,  
authoritative, citable.

#### DERIVATIVE PROCESSING

Vectors + AI responses

Entirely regenerable, rebuilt  
from unchanged source when  
models improve.

#### EDITORIAL

Curator interpretation

Flows one way, archive →  
visitor. Never writes back.

### The stack, all on-device

#### SEARCH

Qdrant + local embeddings

#### REASONING

Local LLM (Qwen), five-layer  
grounding

#### VOICE

Whisper + MeloTTS,  
multilingual

#### ACCESS

CORS-locked Cloudflare  
tunnel

## Six technical contributions

- **Vector embeddings for heritage discovery**, semantic search without depending on controlled vocabulary.
- **Provenance-tracked interpretation**, every answer records its derivation chain back to source.
- **Layered architecture**, permanent heritage held apart from regenerable AI processing.
- **Conversational objects (AUXIO)**, items speaking in the first person from their own validated metadata.
- **Five-layer hallucination prevention**, responses stay grounded under adversarial conditions.
- **Transferable hardware reference**, a self-deployable stack for the GLAM sector.

Live today: ~1,400 public AUXIO pages across institutions on five continents; 3,000+ staff-searchable records from twenty open sources; every object open-licensed, rights-restricted media excluded at ingest. Deployment is a **one-time hardware capital cost, owned outright, not a per-seat cloud subscription.**

## Building and improving it, sovereignly

Two FAMTEC tools underpin how ARCHAI is built and kept current without loosening the sovereignty it promises. **Huggle** is a local coordination layer that lets several AI coding agents and the developer work on the system together, leaving a transparent, auditable development log. It makes continuous improvement tractable and turns the build history itself into research evidence, the commit log as methodology. **Keytec** is a sovereign secrets and API-key wallet: credentials never sit in the open or pass to third parties, every key stays under controlled, auditable custody, and it gives a clean, logged boundary for the rare moments the system needs to reach an external service.

---

## 04 A research agenda for the museum

Seven Research-through-Design experiments, each a working prototype tested in a real institution, mapping directly onto the questions above.

### 1 · Curatorial collection intelligence STAFF-FACING

*How does conversational + semantic search change how curators find connections and build exhibitions?*

Give Collections and Exhibitions staff ARCHAI's staff mode over a scoped set; measure new connections surfaced, time-to-find, and real use in exhibition development. The least-served, most-wanted, lowest-risk place to start.

### 2 · Accessibility experiments VISITOR-FACING

*What access gains come from conversational, multilingual, voice-first interpretation?*

Multilingual AUXIO, voice in/out, plain-language and reading-level adaptation, cognitive-load pacing, high-contrast low-vision modes. The strongest external-funding fit, it extends existing museum accessibility and digital-practice research.

### 3 · Local-LLM evaluation → the sovereign appliance TECHNICAL

*Which local models sustain grounded, hallucination-free, fast interpretation on museum-owned hardware?*

Benchmark local models on institutional records against the five-layer grounding framework, fidelity, latency, power, and publish a hardware reference specification.

### 4 · E-paper signage + NFC PHYSICAL

*Can low-power e-paper + NFC turn a static label into living, updatable, conversational interpretation?*

E-paper labels that redraw from the record (always-on, sustainable); NFC tap → an AUXIO conversation on the visitor's own phone, no app. Prototyped in-gallery.

## 5 · The federated sovereign index FLAGSHIP

*What becomes possible when all open-licensed collections are federated into one local sovereign stack?*

Scale the index into a "connecting archives" brain, cross-collection discovery, provenance across sources, sector-wide questions, held locally and owned outright. The largest ambition, and a natural ARC Linkage-scale project.

## 6 · The living archive COMMUNITY PROVENANCE

*Can moderated visitor contributions become verified archival metadata without eroding authority?*

A public message wall gated by an on-device safety layer (crisis routed to human help, hate held for review); verified facts earn a ★ **green star** and are added to the object's record by curators in the staff app. Community knowledge, feeding the archive.

## 7 · Cultural protocols as architecture GOVERNANCE

*How can cultural protocols be enforceable properties of the system, not just written policy?*

Co-design a cultural-sensitivity layer with the institution's First Nations governance, see the next section, which this experiment leads.

---

# 05 First Nations collections & data sovereignty

Linking First Nations collections to *commercial cloud* AI is precisely what Indigenous Data Sovereignty frameworks forbid, it moves the material offshore and out of community control. A sovereign, local-first system is the opposite by design, and this is where ARCHAI's "ethics as architecture" thesis is tested most seriously.

## The systems

- **Mukurtu CMS**, Indigenous-led, open-source, with differential access and Traditional Knowledge narratives in the metadata.

- **Traditional Knowledge (TK) Labels + Local Contexts Hub**, the protocol/rights layer; communities author the labels and control their delivery.
- **Ara Irititja** and **AIATSIS**, Australian community and national heritage systems.

## How ARCHAI would engage, and the line it holds

ARCHAI can **read and enforce TK Labels**: exclude restricted material at ingest, honour differential access, foreground community narratives, and never write AI output back into the record. Running locally, the data never leaves, satisfying **Authority to Control** and infrastructure sovereignty (CARE Principles; Maiam nayri Wingara).

This can only be a **community-governed partnership, consent-first**. Authority to Control sits with communities. Not everything is linkable, and some material must never be ingested or surfaced. ARCHAI's role is to *respect and enforce* protocols; communities decide. The proposition is not "ingest Indigenous collections", it is "sovereign infrastructure that enforces *your* protocols, on *your* terms, on hardware you control."

## 06 Working together

**RMIT is already ACMI's Major University Partner**, a decade-long relationship, renewed in 2026, spanning the ACMI + RMIT Audience Lab, the ARC-supported Museum Digital Social Futures project, RMIT's Centre of Digital Ecosystems, and an existing ARC Fellowship. ARCHAI extends that partnership into sovereign AI research. Pathways:

- **Scoped pilot**, a small, low-risk, ACMI-approved record set; rights, terminology and cultural protocols set by your team.
- **ARC Linkage Project**, RMIT + ACMI as partners; Major-Partner status strengthens the bid.
- **Audience Lab experiments**, in-gallery prototypes (e-paper/NFC, the living archive) tested with real audiences.
- **Visitor studies & evaluation, co-authored outputs**, and **student WIL projects** extending the existing pipeline.

## What we'd explore together

- ? Which collection subset makes the best low-risk pilot?

- ? Which CMS / DAMS should ARCHAI read from, and what does the API expose?
- ? What rights, licensing and First Nations governance must be enforceable from day one?
- ? A light informal pilot, or a formal research partnership / ARC Linkage bid?
- ? Could the museum host visitor studies to evaluate the living-archive model?

---

Rob Graham · Creative technologist · Founder, Fine Art Media (FAMTEC) · PhD Researcher, RMIT University

rob@fineartmedia.tech · +61 409 336 251 · [fineartmedia.tech/archai](https://fineartmedia.tech/archai)

Try the concept (best on a phone): [fineartmedia.tech/acmi](https://fineartmedia.tech/acmi)

Sources on First Nations frameworks: Mukurtu CMS & TK Labels ([mukurtu.org](https://mukurtu.org)) · Local Contexts ([localcontexts.org](https://localcontexts.org)) · Ara Irititja ([irititja.com](https://irititja.com)) · CARE Principles ([gida-global.org](https://gida-global.org)) · Maiam nayri Wingara ([maiamnayriwingara.org](https://maiamnayriwingara.org)). An independent proposal, not affiliated with or endorsed by any institution named.